25 An alternating voltage of 230 kV is supplied to a transformer, which steps the voltage down to a number of streetlamps with a rating of 230 V 150 W. If the maximum allowable current through the primary coil is 10 A, what is the maximum number of streetlamps that can be connected in parallel?

A 1500 **B** 6500 **C** 10000 **D** 15000

$N_P/N_S = V_P/V_S = 1000 = I_S/I_P \rightarrow I_S = 1000 \times 10 = 1\times10^4$ A

Current in each lamp = P / V = 150 / 230 = 0.65 A

No of lamps = 1×10^4 / 0.65 = 1.5×10^4

26 A 0.31 mW beam of photons with energy 3.11 eV each is incident on a clean metal plate M. The